spatial directions. Figure 8.7 illustrates an example with two extra dimensions that are curled up into the surface of a ball—that is, a sphere. As in the case of the single circular dimension, these extra dimensions are tacked on to every point of the familiar extended dimensions. (For visual clarity we again have drawn only an illustrative sample of the spherical dimensions at regularly spaced grid points in the extended dimensions.) Beyond proposing a different number of extra dimensions, one can also imagine other shapes for the extra dimensions. For instance, in Figure 8.8 we illustrate a possibility in which there are again two extra dimensions, now in the shape of a hollow doughnut—that is, a torus. Although they are beyond our ability to draw, more complicated possibilities can be imagined in which there are three, four, five, essentially any number of extra spatial dimensions, curled up into a wide spectrum of exotic shapes, The essential requirement, again, is that all of these dimensions have a spatial extent smaller than the smallest length scales we can probe, since no experiment has yet revealed their existence,

The most promising of the higher-dimensional proposals were those that also incorporated supersymmetry. Physicists hoped that the partial cancelling of the most severe quantum fluctuations, arising from the pairing of superpartner particles, would help to soften the hostilities between gravity and quantum mechanics. They coined the name higher-dimensional supergravity to describe those theories encompassing gravity, extra dimensions, and supersymmetry.

As had been the case with Kaluza’s original attempt, various versions of higher-dimensional supergravity looked quite promising at first. The new equations resulting from the extra dimensions were strikingly reminiscent of those used in the description of electromagnetism, and the strong and the weak forces. But detailed scrutiny showed that the old conundrums persisted. Most importantly, the pernicious short-distance quantum undulations of space were lessened by supersymmetry, but not sufficiently to yield a sensible theory. Physicists also found it difficult to find a single, sensible, higher-dimensional theory incorporating all features of forces and matter.7 It gradually became clear that bits and pieces of a unified theory were surfacing, but that a crucial element capable of tying them all together in a quantum-mechanically consistent manner was